LAB-ASSIGNMENT : 3

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DDL Questions (Data Definition Language)

- Create the **TEACHER** table with the following attributes: (Name, DeptNo, Date_of_Joining, DeptName, Location, Salary).

```
mysql> CREATE TABLE TEACHER (
    ->     Name VARCHAR(50),
    ->     DeptNo INT,
    ->     Date_of_Joining DATE,
    ->     DeptName VARCHAR(50),
    ->     Location VARCHAR(50),
    ->     Salary DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.18 sec)
```

- Create the **SALES** table with the following attributes: (Sales_No, Salesname, Branch, Salesamount, DOB).

```
mysql> CREATE TABLE SALES (
    ->     Sales_No INT,
    ->     Salesname VARCHAR(50),
    ->     Branch VARCHAR(50),
    ->     Salesamount DECIMAL(10,2),
    ->     DOB DATE
    -> );
Query OK, 0 rows affected (0.05 sec)
```

- Create the **DEPARTMENTS** table with the following attributes: (dept_no, dept_name, dept_location).

```
mysql> CREATE TABLE DEPARTMENTS (
    ->     dept_no INT PRIMARY KEY,
    ->     dept_name VARCHAR(50),
    ->     dept_location VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.04 sec)
```

- Create the **EMPLOYEES** table with the following attributes: (emp_id, emp_name, emp_salary, dept_no).

```
mysql> CREATE TABLE EMPLOYEES (
    ->     emp_id INT PRIMARY KEY,
    ->     emp_name VARCHAR(50),
    ->     emp_salary DECIMAL(10,2),
    ->     dept_no INT,
    ->     FOREIGN KEY (dept_no) REFERENCES DEPARTMENTS(dept_no)
    -> );
Query OK, 0 rows affected (0.08 sec)
```

DML Questions (Data Manipulation Language)

From TEACHER Table

7. Insert five records into the **TEACHER** table.

```
mysql> INSERT INTO TEACHER VALUES
    -> ('Amit', 1, '2018-06-12', 'Mathematics', 'Delhi', 40000),
    -> ('Riya', 2, '2019-07-15', 'Commerce', 'Mumbai', 35000),
    -> ('Sohan', 1, '2017-08-20', 'Mathematics', 'Delhi', 45000),
    -> ('Neha', 3, '2020-01-10', 'Science', 'Chennai', 38000),
    -> ('Arjun', 2, '2016-03-18', 'Commerce', 'Mumbai', 42000);
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

8. Give an increment of 25% salary to teachers working in the Mathematics department.
9. Give an increment of 15% salary to teachers working in the Commerce department.

```
mysql> UPDATE TEACHER
    -> SET Salary = Salary + (Salary * 0.25)
    -> WHERE DeptName = 'Mathematics';
Query OK, 2 rows affected (0.03 sec)
Rows matched: 2  Changed: 2  Warnings: 0

mysql> UPDATE TEACHER
    -> SET Salary = Salary + (Salary * 0.15)
    -> WHERE DeptName = 'Commerce';
Query OK, 2 rows affected (0.00 sec)
Rows matched: 2  Changed: 2  Warnings: 0
```

From SALES Table

- Insert five records into the **SALES** table.

```
mysql> INSERT INTO SALES VALUES
    -> (1, 'Raj', 'North', 50000, '1995-12-21'),
    -> (2, 'Simran', 'South', 45000, '1994-05-10'),
    -> (3, 'John', 'North', 60000, '1992-12-05'),
    -> (4, 'Kiran', 'East', 30000, '1993-07-19'),
    -> (5, 'Meera', 'South', 70000, '1996-12-11');
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

- Calculate the total sales amount in each branch.

```
mysql> SELECT Branch, SUM(Salesamount) AS Total_Sales
    -> FROM SALES
    -> GROUP BY Branch;
+--------+-------------+
| Branch | Total_Sales |
+--------+-------------+
| North  |   110000.00 |
| South  |   115000.00 |
| East   |    30000.00 |
+--------+-------------+
3 rows in set (0.02 sec)
```

- Calculate the average sales amount in each branch.

```
mysql> SELECT Branch, AVG(Salesamount) AS Average_Sales
    -> FROM SALES
    -> GROUP BY Branch;
+--------+---------------+
| Branch | Average_Sales |
+--------+---------------+
| North  |  55000.000000 |
| South  |  57500.000000 |
| East   |  30000.000000 |
+--------+---------------+
3 rows in set (0.00 sec)
```

- Display the Salesname and DOB of salesmen born in the month of December with date format like *21-Dec-09*.

```
mysql> SELECT Salesname,
    -> DATE_FORMAT(DOB,'%d-%b-%y') AS DOB
    -> FROM SALES
    -> WHERE MONTH(DOB) = 12;
+-----------+-----------+
| Salesname | DOB       |
+-----------+-----------+
| Raj       | 21-Dec-95 |
| John      | 05-Dec-92 |
| Meera     | 11-Dec-96 |
+-----------+-----------+
3 rows in set (0.00 sec)
```

- Display the Salesname and DOB of salesmen in alphabetical order of the month.

```
mysql> SELECT Salesname, DOB
    -> FROM SALES
    -> ORDER BY MONTHNAME(DOB);
+-----------+------------+
| Salesname | DOB        |
+-----------+------------+
| Raj       | 1995-12-21 |
| John      | 1992-12-05 |
| Meera     | 1996-12-11 |
| Kiran     | 1993-07-19 |
| Simran    | 1994-05-10 |
+-----------+------------+
5 rows in set (0.00 sec)
```

From JOIN Operations

- Display employee details and department details where the department numbers are the same in both tables.

```
mysql> SELECT *
    -> FROM EMPLOYEES E
    -> INNER JOIN DEPARTMENTS D
    -> ON E.dept_no = D.dept_no;
+--------+----------+------------+---------+---------+-----------+---------------+
| emp_id | emp_name | emp_salary | dept_no | dept_no | dept_name | dept_location |
+--------+----------+------------+---------+---------+-----------+---------------+
|    101 | Rahul    |   40000.00 |       1 |       1 | HR        | Delhi         |
|    105 | Rohit    |   38000.00 |       1 |       1 | HR        | Delhi         |
|    102 | Sneha    |   50000.00 |       2 |       2 | Finance   | Mumbai        |
|    104 | Anita    |   55000.00 |       2 |       2 | Finance   | Mumbai        |
|    103 | Vikas    |   45000.00 |       3 |       3 | IT        | Chennai       |
+--------+----------+------------+---------+---------+-----------+---------------+
5 rows in set (0.00 sec)
```

- Display the employee name and department name using a LEFT OUTER JOIN.

```
mysql> SELECT E.emp_name, D.dept_name
    -> FROM EMPLOYEES E
    -> LEFT JOIN DEPARTMENTS D
    -> ON E.dept_no = D.dept_no;
+----------+-----------+
| emp_name | dept_name |
+----------+-----------+
| Rahul    | HR        |
| Sneha    | Finance   |
| Vikas    | IT        |
| Anita    | Finance   |
| Rohit    | HR        |
+----------+-----------+
5 rows in set (0.01 sec)
```

- Display the employee name and department name using a RIGHT OUTER JOIN.

```
mysql> SELECT E.emp_name, D.dept_name
    -> FROM EMPLOYEES E
    -> RIGHT JOIN DEPARTMENTS D
    -> ON E.dept_no = D.dept_no;
+----------+-----------+
| emp_name | dept_name |
+----------+-----------+
| Rahul    | HR        |
| Rohit    | HR        |
| Sneha    | Finance   |
| Anita    | Finance   |
| Vikas    | IT        |
+----------+-----------+
5 rows in set (0.00 sec)
```

- Display the details of employees who draw a salary greater than the average salary.

```
mysql> SELECT emp_name, emp_salary
    -> FROM EMPLOYEES
    -> WHERE emp_salary > (SELECT AVG(emp_salary) FROM EMPLOYEES);
+----------+------------+
| emp_name | emp_salary |
+----------+------------+
| Sneha    |   50000.00 |
| Anita    |   55000.00 |
+----------+------------+
2 rows in set (0.02 sec)
```

TCL Questions (Transaction Control Language)

- Perform the **ROLLBACK** command on the TEACHER table.
- Perform the **COMMIT** command on the TEACHER table.
- Write a query to implement a **SAVEPOINT**.

```
mysql> rollback;
Query OK, 0 rows affected (0.00 sec)

mysql> commit;
Query OK, 0 rows affected (0.00 sec)

mysql> savepoint a;
Query OK, 0 rows affected (0.00 sec)
```

DCL Questions (Data Control Language)

- Write a query to **grant all privileges** on the EMPLOYEES table to DEPARTMENTS.
- Write a query to **grant selected privileges** on the EMPLOYEES table to DEPARTMENTS.
- Write a query to **revoke all privileges** on the EMPLOYEES table from DEPARTMENTS.
- Write a query to **revoke selected privileges** on the EMPLOYEES table from DEPARTMENTS.

```
mysql> GRANT ALL PRIVILEGES
    -> ON EMPLOYEES
    -> TO 'dept_user'@'localhost';
Query OK, 0 rows affected (0.03 sec)

mysql> GRANT SELECT, INSERT
    -> ON EMPLOYEES
    -> TO 'dept_user'@'localhost';
Query OK, 0 rows affected (0.04 sec)

mysql> REVOKE ALL PRIVILEGES
    -> ON EMPLOYEES
    -> FROM 'dept_user'@'localhost';
Query OK, 0 rows affected (0.03 sec)
```